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204I – DATA STRUC

1. **Understanding SQL Joins**

MEANING:

- SQL JOIN clauses are used to combine rows from two or more tables in a relational database based on a related column between them.

PURPOSE:

- To retrieve related data stored in different tables
- To avoid data duplication
- To create meaningful results from multiple sources

EXAMPLE:

- Having Student table and a Courses Table, you can use a JOIN to show which student is enrolled in which course.

2. **CREATING A DIAGRAM**

TABLE FOR STUDENTS

Results | Messages

	id	name
1	1	Ana
2	2	Ben
3	3	Cara

TABLE FOR COURSES

100 % 1

Results | Messages

	id	course
1	1	Math
2	2	Science
3	4	English

1. **INNER JOIN**
   - Only matching IDs
   Result: 1, 2

   2. LEFT JOIN
   - All from left + matches
   Result: 1, 2, 3

   3. RIGHT JOIN
   - All from right + matches
   Result: 1, 2, 4

   4. FULL OUTER JOIN
   - All records from both
   Result: 1, 2, 3, 4

3. PROVIDING EXAMPLES

TABLE FOR STUDENTS

Results | Messages

	id	name
1	1	Ana
2	2	Ben
3	3	Cara

TABLE FOR COURSES

100 % 1

Results | Messages

	id	course
1	1	Math
2	2	Science
3	4	English

INNER JOIN

SQLQuery1.s...\Yasy (77))*

```sql
SELECT Students.id, Students.name, Courses.course
FROM Students
INNER JOIN Courses
ON Students.id = Courses.id;
```

100 % 7 0

Results | Messages

	id	name	course
1	1	Ana	Math
2	2	Ben	Science

- The INNER JOIN query retrieves only the records where the id in the Students table matches the id in the Courses table, meaning only students who have a corresponding course are included in the result.

LEFT JOIN

SQLQuery1.s...\Yasy (77))*

```sql
SELECT Students.id, Students.name, Courses.course
FROM Students
LEFT JOIN Courses
ON Students.id = Courses.id;
```

100 % 7 0

Results | Messages

	id	name	course
1	1	Ana	Math
2	2	Ben	Science
3	3	Cara	NULL

- The LEFT JOIN query retrieves all records from the Students table and matches them with the Courses table, showing NULL values for students who do not have a corresponding course.

RIGHT JOIN

SQLQuery1.s...\Yasy (77))*

```sql
SELECT Students.id, Students.name, Courses.course
FROM Students
RIGHT JOIN Courses
ON Students.id = Courses.id;
```

100 % 7 0

Results | Messages

	id	name	course
1	1	Ana	Math
2	2	Ben	Science
3	NULL	NULL	English

- The RIGHT JOIN query retrieves all records from the Courses table and matches them with the Students table, showing NULL values for courses that do not have a corresponding student.

FULL OUTER JOIN

```sql
SELECT Students.id, Students.name, Courses.course
FROM Students
FULL OUTER JOIN Courses
ON Students.id = Courses.id;
```

100 % 7 0

Results Messages

	id	name	course
1	1	Ana	Math
2	2	Ben	Science
3	3	Cara	NULL
4	NULL	NULL	English

- The FULL OUTER JOIN query retrieves all records from both the Students and Courses tables, showing NULL values wherever there is no match between the two tables.

4. CONCLUSION

SQL joins are important in database management because they help combine and analyze data from different tables easily. Without using joins, it would be harder and more repetitive to get related information.

In this activity, I learned how the different types of joins work and when to use them. For example, the INNER JOIN is used when we only want matching data from both tables, while the LEFT JOIN is useful when we want to include all records from one table even if there is no match.

At first, I found it confusing to understand the difference between LEFT JOIN and RIGHT JOIN, but after practicing some examples, I was able to understand it better. Overall, SQL joins are very useful in real-life situations like student systems, business reports, and inventory management.